1.18 Solution: (a) If $G(\mathbf{x}, \mathbf{x}')$ is the Green function in the absence of surface S_1 with other surfaces held at zero potential, the potential in the whole space exterior to the surfaces is given by, of course,

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \oint_{S_1} \sigma_1(\mathbf{x}')G(\mathbf{x}, \mathbf{x}')da',$$

and the electrostatic energy is

$$W = \frac{1}{2} \oint_{S_1} \sigma(\mathbf{x})\Phi(\mathbf{x})da = \frac{1}{8\pi\epsilon_0} \oint_{S_1} da \oint_{S_1} da' \; \sigma_1(\mathbf{x})G(\mathbf{x}, \mathbf{x}')\sigma_1(\mathbf{x}').$$

(b) For any trial surface charge density distribution on surface S_1, we have

$$W = \frac{Q^2}{2C} = \frac{1}{2C} \left[\oint \sigma(\mathbf{x})da \right]^2,$$

we can write the capacitance as

$$C^{-1}[\sigma] = \frac{\oint_{S_1} da \oint_{S_1} da' \; \sigma(\mathbf{x})G(\mathbf{x}, \mathbf{x}')\sigma(\mathbf{x}')}{4\pi\epsilon_0 \left[\oint \sigma(\mathbf{x})da \right]^2}.$$

The variation with respect to σ is

$$\begin{aligned} 4\pi\epsilon_0 \frac{\delta C^{-1}[\sigma]}{\delta\sigma} &= \frac{2 \oint_{S_1} da' G(\mathbf{x}, \mathbf{x}')\sigma(\mathbf{x}')}{Q^2} - \frac{2 \oint_{S_1} da \oint_{S_1} da' \; \sigma(\mathbf{x})G(\mathbf{x}, \mathbf{x}')\sigma(\mathbf{x}')}{Q^3} \\ &= \frac{2}{Q^3} \left[Q\Psi(\mathbf{x}) - \oint_{S_1} \sigma(\mathbf{x})\Psi(\mathbf{x})da \right], \end{aligned}$$

where

$$\Psi(\mathbf{x}) = \oint_{S_1} da' G(\mathbf{x}, \mathbf{x}')\sigma(\mathbf{x}')$$

is the potential calculated from the trial surface change density. For true surface charge density $\sigma_1(\mathbf{x})$, the surface is equipotential and $\Psi(\mathbf{x})$ will be a constant on the surface. Therefore,

$$\left. \frac{\delta C^{-1}[\sigma]}{\delta\sigma} \right|_{\sigma_1} = 0,$$

i.e., $C^{-1}[\sigma]$ is stationary at σ_1. By Thomson's theorem, the true surface charge density will lead to the minimum electrostatic energy, that is to say, $C^{-1}[\sigma]$ is the lowest for σ_1. Then, any trial surface charge density gives higher C^{-1}, or lower capacitance. Therefore, the true capacitance is bounded from below by the capacitance calculated from an arbitrary trial surface charge density.